LECTURE 4

TYPED BY AOWEN FAN

In this talk, we collect main results of Section 4 of [BSY22], which will offer a main tool in our next talk. We discuss the nilpotence detection in a locally rigid category. Moreover, we will apply nilpotence detecting techniques to Lubin-Tate-tilt adjunction and give a criterion for a ring to be perfect of Krull dimension 0.

1. Locally rigid categories

Definition 1.1. We use $\mathrm{Pr}_L^{\mathrm{rig}}$ to denote the category of locally rigid compactly generated categories, with morphisms the strongly continuous functors.

The following lemmas are fundamental.

Lemma 1.2. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$, then*

(1) $c \in \mathcal{C}^\omega$ implies $c^\vee \in \mathcal{C}^\omega$.
(2) Suppose we have $a \in \mathcal{C}$ such that $a \otimes c = 0$ for any $c \in \mathcal{C}^\omega$, then $a = 0$.

Lemma 1.3. *Consider $\mathrm{Mod}_B(\mathcal{C})$ for $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$ and $B \in \mathrm{CAlg}(\mathcal{C})$. Then*

(1) $\mathrm{Mod}_B(\mathcal{C})$ is in $\mathrm{Pr}_L^{\mathrm{rig}}$.
(2) $- \otimes B : \mathcal{C} \to \mathrm{Mod}_B(\mathcal{C})$ is strongly continuous.

2. Nilpotence and detecting nilpotence

Definition 2.1. Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$. Suppose $f : a \to b$ is a morphism in $\mathcal{C}$ and $e \in \mathcal{C}$.

(1) We say that f is nilpotent at e, if there is some $m \in \mathbb{N}$ such that $f^{\otimes m} \otimes \mathrm{id} : a^{\otimes m} \otimes e \to b^{\otimes m} \otimes e$ is null.
(2) We say f is nilpotent if it is nilpotent at $\mathbb{1}_{\mathcal{C}}$.
(3) We say f is locally nilpotent if it is nilpotent at every compact object.

The following lemma is immediate.

Lemma 2.2. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$.*

(1) Suppose we have $f : c \to a$ with c compact, then f is locally nilpotent if and only if f is nilpotent.
(2) Suppose d is dualizable. Then $f : d \otimes a \to b$ is nilpotent at e if and only if $f^\vee : a \to b \otimes d^\vee$ is nilpotente at e.
(3) Suppose d is dualizable, then f is nilpotent at d if and only if $f \otimes d$ is nilpotent.

Definition 2.3. Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{rig}}$, $T \in \mathcal{C}$ and $c \in \mathcal{C}^\omega$.

(1) We say T detects nilpotence at c if for every map $f : c \to a$, f is nilpotence at T if and only if it is nilpotence.
(2) We say T detects nilpotence if it detects nilpotence at every compact object.

In reality, we will deal with the weak rings, that means, objects T in $\mathcal{C}$ that with a unit map $\mu_T : \mathbb{1}_{\mathcal{C}} \to T$ such that $\mu_T \otimes T$ admits a retract.

Lemma 2.4. *Suppose T_1, T_2 are two weak rings. Then T_1, T_2 detect nilpotence imply $T_1 \otimes T_2$ detects nilpotence.*

Proposition 2.5. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$, $T \otimes -$ is conservative, then T detects nilpotence.*

Example 2.6. Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$ and $e \xrightarrow{x} \mathbb{1}_{\mathcal{C}}$ with $e \in \mathrm{Pic}(\mathcal{C})$. Then $\mathbb{1}_{\mathcal{C}}[x^{-1}] \oplus \mathbb{1}_{\mathcal{C}}/x$ detects nilpotence.

Example 2.7. Let $R \in \mathrm{CAlg}_{E(k)}^{\wedge}$. Since each v_i is locally nilpotence, then $R//\mathfrak{m} = R/v_1 \otimes_R \cdots \otimes_R R/v_{n-1}$ detects nilpotence in $\mathrm{Mod}_R^{\wedge}$.

Proposition 2.8. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$.*

1. *Suppose that $T \in \mathcal{C}$ is a weak ring, then T detects nilpotence, then $B \otimes T$ detects nilpotence in $\mathrm{Mod}_B(\mathcal{C})$.*
2. *Filtered colimits of nilpotence detecting weak rings are nilpotence detecting weak rings.*

The key results for nilpotence detecting theory is the following.

Theorem 2.9. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$. Then nilpotence detecting morphisms in $\mathrm{CAlg}(\mathcal{C})$ form a weakly saturated collection.*

Recall that we say $f : C \to D$ has right lifting property with respect to $X \in \mathcal{C}$ if there is always a solution

$$\begin{array}{ccc} C & \longrightarrow & X \\ {\scriptstyle f}\downarrow & \nearrow & \\ D & & \end{array}$$

In this case, we denote $f \perp X$.

By Quillen's small objects argument, we can prove the following.

Theorem 2.10. *In $\mathrm{CAlg}(\mathcal{C})$:*

1. *$\mathbb{1}[t] \to \mathbb{1}[t^{\pm 1}] \times \mathbb{1}$ detects nilpotence.*
2. *$\mathbb{1}[t^{1/p^\infty}] \to \mathbb{1}[t^{\pm 1/p^\infty}] \times \mathbb{1}$ detects nilpotence.*

3. Apply to perfect rings of Krull dimension 0

We have the following criterion for Krull dimension 0 commutative rings.

Proposition 3.1. *Let B be an ordinary commutative ring, then the followings are equivalent.*

1. *B is reduced of Krull dimension 0*
2. *We have*

$$(\mathbb{Z}[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{Z}[t^{\pm 1}] \times \mathbb{Z}) \perp B$$

Similarly, if we restrict ourselves to perfect rings, we have

Proposition 3.2. *Let A be a perfect ring, then the followings are equivalent.*

1. *A is reduced of Krull dimension 0*
2. *We have*

$$(\mathbb{F}_p[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{F}_p[t^{\pm 1}] \times \mathbb{F}_p) \perp B$$

Corollary 3.3. *Suppose B is an $\mathbb{F}_p$-algebra such that $B^\flat$ is of Krull dimension 0, then the natural map $B^\flat \to B$ is injective.*

One can apply the Lubin-Tate-tilt adjunction. We have

Corollary 3.4. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$ be p-complete. In $\mathrm{CAlg}(\mathcal{C})$, consider*

$$f : \mathbb{1}_{\mathcal{C}}[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{1}_{\mathcal{C}}[t^{\pm 1}] \times \mathbb{1}_{\mathcal{C}}$$

Then $f \perp R$ if and only if $R^\flat \in \mathrm{Perf}_p$ is of Krull demension 0.

Theorem 3.5. *Let $\mathcal{C} \in \mathrm{Pr}_L^{\mathrm{st}}$ be p-complete. Recall that we have a spherical-tilt adjunction*

$$\mathbb{W}_{\mathcal{C}}(-) : \mathrm{Perf}_p \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

Let $f : A \to B$ be a morphism in Perf_p. Then the following are equivalent

(1) $\mathbb{W}_{\mathcal{C}}(B) \in \mathrm{Mod}_{\mathbb{W}_{\mathcal{C}}(A)}$ is conservative
(2) The map $\mathbb{W}_{\mathcal{C}}(A) \to \mathbb{W}_{\mathcal{C}}(B)$ detects nilpotence.
(3) The base-change functor $\mathrm{Mod}_{\mathbb{W}_{\mathcal{C}}(A)} \to \mathrm{Mod}_{\mathbb{W}_{\mathcal{C}}(B)}$ is nil-conservative.
(4) $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is surjective.

Conclusions

(1) Detecting nilpotence means that the corresponding maps between Spec are surjective.
(2) $(\mathbb{1}_{\mathcal{C}}[t] \xrightarrow{(t\mapsto t)\times(t\mapsto 0)} \mathbb{1}_{\mathcal{C}}[t^{\pm 1}] \times \mathbb{1}_{\mathcal{C}})R$ means R is of Krull dimension 0. By small objects argument, this gives a cover

$$\mathrm{Spec}(\prod_{A\to k} k) \to \mathrm{Spec}(A)$$

We will show that under some conditions the former can be written as the spectrum of a Lubin-Tate theory in our next talk.

References

[BSY22] Robert Burklund, Tomer M Schlank, and Allen Yuan. "The chromatic nullstellensatz". In: *arXiv preprint arXiv:2207.09929* (2022) (cit. on p. 1).